Biological Measure	**Reference/range/threshold**	**Source**
Heart Rate	60-100 bpm is normal. BPM ≥120 is an indicator of stress in a study on VR and Arachnophobia Another reference: HRz= (HR current minus mean of personal baseline HR)/SD of personal baseline) HRz = ≥ +1.0 and sustained ≥10 s is an elevated physiological arousal cue.	https://www.mayoclinic.org/healthy-lifestyle/fitness/expert-answers/heart-rate/faq-20057979 https://doi.org/10.1080/08995605.2021.1897494 https://doi.org/10.3390/healthcare13202617
Heart Rate Variability - RMSSD	Normal range: 42 ± 15 ms over a 5-minute resting window. Outside this range is abnormal.	https://doi.org/10.3389/fpubh.2017.00258 https://doi.org/10.3390/s24124003
Heart Rate Variability - SDNN	52.0 ± 19.6 ms over a 5-minute resting window. Outside this range is abnormal.	https://doi.org/10.3390/s24124003
Heart Rate Variability - LF/HF Ratio	Normal: ≈2.8 ± 2.6 over a 5-minute resting window.	https://doi.org/10.13140/RG.2.2.29460.00645
Blood Pressure - Systolic and Diastolic	Normal systolic BP <120 mmHg and diastolic BP <80 mmHg Elevated systolic BP is 120-129 Diastolic BP greater than 80 is abnormal	https://doi.org/10.1016/j.jacc.2025.05.007
Skin Conductance Response	Threshold of 0.05 microsiemens is commonly used to identify SCRs. SCR typical value for at rest: 1-3 SCRs per minute	https://public-pages-files-2025.frontiersin.org/articles/1432180/file/supplementary_file_2.pdf/1432180_supplementary-file_2/1

	Stress: >8 responses/min or SCR frequency >20 per minute indicates high arousal.	https://www.birmingham.ac.uk/documents/college-les/psych/saal/guide-electrodermal-activity.pdf
Skin Conductance Level	Sustained SCL of ≈2× the patient's baseline is indicative of elevated physiological arousal. Typical ranges is between 2-16 microsiemens at rest	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0305864 https://www.birmingham.ac.uk/documents/college-les/psych/saal/guide-electrodermal-activity.pdf
Respiratory Rate	Autonomic (resting) breathing rate 0.20–0.25 Hz = 12–15 bpm "Emotional" breathing rate approximately 0.32 Hz = 19 bpm Or >30 breaths/min sustained	https://pmc.ncbi.nlm.nih.gov/articles/PMC11025454/
Tidal Volume	Tidal volume = Around 500 ml is normal in a healthy adult male, 400 ml in a healthy female. Above 10ml/kg of IBW is indicative of high arousal	https://www.ncbi.nlm.nih.gov/books/NBK482502/
Voice f0 in hz See notes at the end of the document.	+5% pitch compared to baseline = stress cue **Typical mean F0 at rest: ≈220 Hz women ≈130 Hz men	https://doi.org/10.1177/0301006620978378 ** This is a preprint, not a peer-reviewed source: https://doi.org/10.48550/arXiv.2207.08534
Speech rate	+8% speech rate compared to baseline = stress cue	https://doi.org/10.1177/0301006620978378
Head movement	5-s rolling standard deviation of head-pose velocity (yaw,pitch, and roll combines), z-scored to the first 2 minutes of	https://doi.org/10.3390/healthcare13202617

	exposure. z ≥ +1.0 sustained for ≥ 5 s High arousal/stress cue	
Pupils	sustained 10% to 30% increase in baseline pupil diameter = indicator of acute stress	https://doi.org/10.1016/j.cpnec.2026.100370
EEG alpha/beta ratio	Stress: ≤0.60 A ratio around 0.6 or below consistently represented dress conditions in a VR experiment.	https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0305864

- Reference values were not found for eye tracking/pupillometry, speech and voice, respiratory variability, heart rate variability pnn50, and EEG.
- A systematic review has found that over 39 studies, there were no consistent acoustic patterns identified for fear or anxiety. Generally higher voice f0 in Hz average is an indicator of high arousal but there is no specific reference or threshold. In the systematic review, one Public-speaking TSST study found more and longer pauses; an anxiety study also found longer pauses…again no reference or threshold identified.
  Source: https://doi.org/10.1371/journal.pone.0328833